

SAMIR JAGTAP

SDET | Agentic AI QA Architect | MCP-Based Multi-Agent Test Automation
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PROFESSIONAL SUMMARY

SDET with 4.5 years of industry experience (EY, Markgenic) across Wealth Management, Investment Banking, HR, and Telecom domains. Currently specializing in Agentic AI for QA - built 8 production-grade AI-orchestrated frameworks using multi-agent QA pipelines (Claude Terminal + MCP), implementing the same orchestration patterns as LangChain/CrewAI/AutoGen but with MCP's production-ready approach. Configured 12+ MCP servers and completed 390+ hours of certifications. Proficient in Playwright (JS/TS), Selenium (Java), REST Assured, Postman, Jenkins CI/CD, Docker, and Azure DevOps.

TECHNICAL SKILLS

Test Automation	Selenium WebDriver (Java), Playwright (JS/TS), Cucumber BDD, TestNG, POM
API Testing	REST Assured, Postman, Newman, GraphQL, SOAP, OAuth 2.0
AI & Agentic Tools	Claude Terminal, GitHub Copilot (Browser & DB Sub-Agents), MCP Servers (Playwright, Selenium, Jira, MySQL, GitHub, Docker, Azure DevOps), n8n, ChatGPT, GPT4All, Gemini Pro Multi-Agent Orchestration (MCP-based, equivalent to LangChain/CrewAI/AutoGen patterns)
CI/CD & DevOps	Jenkins, Azure DevOps Pipelines, GitHub Actions, Docker, Selenium Grid
Languages & DB	Java, JavaScript, TypeScript, SQL, Unix MySQL, SQL Server
Reporting & Tools	Allure, ExtentReports, Playwright HTML Reporter Jira, HP ALM, Azure Boards Maven, Git
AI Concepts	Prompt Engineering (Zero-shot, Few-shot, CoT), Multi-Agent Collaboration, MCP Protocol

PROFESSIONAL EXPERIENCE

Ernst & Young (EY) | Software Test Engineer Mar 2022 - May 2023 | Bangalore

Wealth Management System | Wealth Asset Management

- Designed 150+ E2E test scenarios with the BA team; achieved 95% test coverage across critical portfolio and asset modules, catching defects before UAT.
- Detected and documented 80+ defects in HP ALM with root-cause analysis, reducing production escape rate and saving estimated 2 sprint cycles of post-release hotfixes.

Leave Management System | Human Resources

- Automated regression suite using POM design pattern with cross-browser execution (Chrome, Firefox, IE); reduced regression cycle time by ~40%, freeing 2 days per sprint for exploratory testing.
- Integrated Selenium framework with Jenkins (Maven) for nightly builds; reviewed 100+ test cases per sprint in Agile ceremonies.

Markgenic Software | Test Engineer Dec 2018 - Feb 2022 | Pune

Equity Trade System | Investment Banking

- Led testing across 3 major releases with 200+ E2E scenarios covering trade execution, settlement, and reconciliation flows; maintained 100% defect closure rate in HP ALM.
- Sustained ~30% defect detection rate during regression cycles, preventing high-severity trade calculation errors from reaching production.

Tele 2 Billing System | Telecom Billing

- Built 120+ test scenarios with full RTM coverage; identified 25+ data integrity defects in billing calculations that would have caused revenue leakage in production.

CAREER DEVELOPMENT & UPSKILLING

AI-QA Research Sabbatical Jun 2023 - Present | Pune

- Built 8 AI-orchestrated automation projects (all on GitHub with documentation) using multi-agent pipelines, MCP servers, Playwright, Selenium, and GitHub Copilot sub-agents.
- Configured 12+ MCP servers (Playwright, Selenium, Jira, MySQL, GitHub, Docker, Azure DevOps) and implemented 18+ specialized AI QA agents for real automation tasks.
- Mastered Agentic AI architecture: multi-agent pipelines, self-healing test frameworks, MCP integration, and Playwright CLI automation.
- Completed 10 certifications (390+ hours) in Automation Testing, AI in Testing, Playwright, BDD, API Testing, and SQL.

KEY PROJECTS (AI-ORCHESTRATED AUTOMATION)

1. EventHub - Playwright BDD Framework | Playwright, Cucumber/Gherkin, TypeScript, GitHub Actions

- Architected a 7-agent AI pipeline (Domain SME, Test Strategist, Automation Developer, Code Reviewer, CI/CD Agent) via Claude Terminal for end-to-end test generation and validation.
- Orchestrated seamless multi-agent workflows by integrating specific MCP servers (Playwright MCP, Excel MCP, GitHub MCP, and Jira MCP).
- Reduced test authoring time by ~80% (51 cross-pyramid scenarios generated); reduced CI selector failures by ~75% via strict locator hierarchy enforcement.

github.com/samirjagtap4030/eventhub_playwright_bdd

2. EventHub - Selenium BDD Framework | Java, Selenium 4, Cucumber BDD 7, TestNG, Jenkins, ExtentReports

- Designed multi-agent automation pipeline featuring specialized agents (/generate-tests, /review-tests, /test-strategy) for live browser-validated test generation.
- Leveraged Selenium MCP and MySQL MCP for database-driven AI validation, achieving ~95% test review consistency through AI code review with [CRITICAL]/[IMPORTANT]/[SUGGESTION] severity tagging.
- Increased parallel throughput by ~300% via 4-agent Jenkins partitioning, cutting regression execution from 120 to 35 minutes.

github.com/samirjagtap4030/eventhub_selenium_bdd

3. OrangeHRM - Agentic Workflow | Excel MCP, Playwright MCP, Jira MCP

- Built an end-to-end agentic workflow: automated test case creation in Excel & Jira -> execution via Playwright MCP -> result synchronization.
- Fully automated the manual testing life cycle using 3 concurrent MCP agents, reducing end-to-end test cycle time by ~70%.

github.com/samirjagtap4030/OrangeHRM_JiraMCP_Manual | [OrangeHRM_JiraMCP_Automation](#)

4. AI Agent MCP - GitHub Copilot Sub-Agents | GitHub Copilot, Browser Agent, Database Agent, MCP

- Built custom Browser Agent and Database Agent as GitHub Copilot sub-agents using the MCP protocol.
- Eliminated manual browser interactions and database querying steps, improving overall QA productivity by ~60%.

github.com/samirjagtap4030/AiAgentMCP

5. Playwright Agents & CLI Automation | Playwright, Planner Agent, Generator Agent, Healer Agent, TypeScript, Node.js

- Utilized Playwright's inbuilt AI agents via Claude Terminal - Planner (app exploration & test planning), Generator (create & execute tests), and Healer (self-healing & auto-repair for failed tests) - demonstrating fully AI-driven test lifecycle.
- Built Playwright CLI automation demo via Claude Terminal - test generation, execution, debugging, and reporting; serves as reference guide for CLI-based automation workflows.

github.com/samirjagtap4030/PlaywrightAgents | github.com/samirjagtap4030/playwrightCliDemo

CORE FRAMEWORK EXPERTISE

Playwright JS/TS

- Implemented POM, BDD (Cucumber), Data-Driven Testing (ExcelJS/JSON), visual testing, network mocking (route.fulfill/abort), and parallel cross-browser execution on Azure Cloud; integrated Playwright MCP for AI-driven test generation.
- Integrated with Jenkins & Azure DevOps Pipelines; configured Allure/HTML reporting with traces, videos, screenshots.

Selenium Java

- Built Hybrid Frameworks with POM, TestNG, Cucumber BDD, Maven; implemented IRetry Analyzer, ThreadLocal for parallel safety, ExtentReports with auto-screenshots on failure; integrated Selenium MCP via Claude Terminal.
- Executed scripts on Selenium Grid (Hub/Node); Jenkins CI integration with parameterized jobs and nightly schedules.

Postman & REST Assured

- Automated REST, SOAP, GraphQL APIs with OAuth 2.0, Bearer Tokens; performed Data-Driven Testing via CSV; JSON Schema validation; Mock Servers; Newman CLI + Jenkins integration.
- Built Cucumber REST Assured frameworks with reusable step definitions, utility files, logging, global properties, and Cucumber HTML reporting.

EDUCATION

BE in Production Engineering - Shivaji University, Maharashtra | 2014

CERTIFICATIONS

Playwright JS/TS Automation Testing from Scratch & Framework (Feb 2026) | Generative AI in Software Testing (Feb 2026) | Selenium WebDriver with Java - Basics to Advanced (Dec 2024) | Cucumber BDD MasterClass with Selenium Framework (Dec 2024) | REST API Testing - Rest Assured Java (Dec 2024) | API Testing using Postman, SoapUI & RestAssured (Dec 2024) | Selenium with Java, Cucumber & Frameworks (Dec 2024) | Manual Software Testing + Agile with Jira (Nov 2023) | Test Management using HP ALM / QC Quality Center (Nov 2023) | SQL + Security Testing (Dec 2024)